

DEPARTMENT OF ELECTRONICS ENGINEERING

**PROJECT TITLE: DESIGN AND ANALYSIS PROJECT OF A REAL TIME
VOTING SYSTEM**

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Declaration by Candidate

I Brave O. Tafireyi declare on behalf of my group that this project report entitled “Design and Analysis Project of a Real-Time Voting System” is our original work and has not been submitted at any other university for the award of any other Degree.

Acknowledgements

I would like to express my gratitude to our Supervisor Mr S. Kanjera for his invaluable patience and feedback throughout the course of the project. I would also like to extend my sincere gratitude to prospective Engineer M. Chada and Engineer F. Mavhunila for the knowledge shared on improvements and valuable insight on the project. Special thanks to my group members who also made it possible for the project to be completed, for their cooperation and dedication throughout the project.

Abstract

This report is a compilation of all the work, relevant information and research that was carried out during the project design. It has been found out that the traditional voting systems used to select certain members of the public into power has become too dependent on human interaction and involvement posing a greater risk of manipulation and rigging of the voting process. The project design aims to minimize human interaction in voting and results calculation as there has been cases where the voters don't match the number of votes which was raising concerns on the credibility of the whole process. The proposed design is a digital system which uses counters and registers to capture votes in real-time and display the winner without need for human counting.

Chapter 1: Introduction

1.1 Background

Democratically people have always chosen to pick an idea based on a decision the majority deems to be right. This has mostly been the case be it in choosing a leader or making a decision. This is done via voting but voting has been done in many various ways and some of which been having a lot of issues. It has been found out that the traditional voting systems used to select certain members of the public into positions has become too dependent on human interaction and involvement posing a greater risk of manipulation and rigging of the voting process.

1.2 Problem Statement

At an institution, a group of people wish to hold a voting process in order to pick a leader between 3 running candidates that the majority is willing to have. There is need to develop a system that can make the voting process be over-gone smoothly by eliminating some of the issues that may be faced if existing voting methods were to be used.

1.2.1 Research Statement

Required to research and design a digital system that captures votes between the three running candidates and displaying the results in real-time. The system to also to display the winner so as to minimize human interaction in counting.

1.3 Objectives

The objectives of this project are to:

- Hold research on the different existing technologies and methodologies used for voting.
- Propose a System to hold the voting process for the Institution.
- Design a Voting System for the Institution.

1.4 Justification

The digital voting system shall reduce the time spent on long queues at polling stations and the possibility of ballot paper shortages as the new digital system will only need the participant to press a button for the candidate of their choice and the process is done whilst there is a 5-10 second window period to cast a vote.

The new system will also ensure an instant announcement of the winner without any counting process. Cases of vote miscounts will also be reduced since at the back-end of this system resides a well-developed database using MYSQL that can provide the correct data once it's correctly queried. Use of digital systems in voting processes also reduces costs incurred in hiring clerks for the success of the manual process.

Chapter 2: Literature Review

2.1 Introduction

Throughout the project research on the history of technologies and methodologies used, components of the systems and advancement of the technology in regards to the technical equipment and systems used in voting processes was undertaken so as to get a broader view of what exactly we need to implement in our voting system, what it does, how the technology works and how it is to be implemented in our system to be designed.

2.1 Existing Technologies

The earliest voting methodology was called ‘*viva voce*’ meaning “with the living voice,” from around the colonial era till early 1800’s [1]. Such a system would depend on the majority of those who raise their hands in vote of something or the loudness of peoples screams on vote [2]. Types of this type of voting system include the Single round plurality, voice vote, raising vote and the two round system[3].



Fig.2.0. Viva voce voting by hand raising.

This type of voting system was unreliable and very prone to errors and bias as estimation was used by just seeing or hearing. Given an instructor was paid off to let one candidate win he or she could easily declare a win for the one he or she wants to win without any proof needed.

In the late 1800’s the first voting machine was invented called the acme voting machine. The Acme Voting Machine was aimed at securing an honest vote. It did so by providing

a way for poll overseers to detect additional voter fraud. The machine allowed poll overseers to count the exact number of ballots cast compared to registered voters. It had a tabulator activated by a lever mechanism that released the ballot into the box.



Fig.2.1. Acme Voting machine of the Late 1800's

A tabulator is a computing machine that reads and summarizes data, such as punched cards, producing lists or totals. Prior to this invention, there was no way of ensuring only one vote per person. Integrity was left in the hands of the voter, who would often swear on the Bible in front of a Judge that they were indeed the person they claimed to be and that it was their first and only vote. This subsequently allowed many voters to cast their vote multiple times. Although the Acme Voting Machine was great at tabulating votes and aided in voter fraud prevention, the integrity of the count was often questionable [1].

Other machines were invented as time progressed such as the Lever style voting machine, the Coyle machine, votomatic punching card system and more. Regardless of these machines being invented and all, people still regularly chose an old school yet effective way of voting called balloting. Balloting is the standard election method for most parliamentary elections that started around time after the Civil war. A candidate is elected if they receive support from a majority of the voters [3]. Balloting is now done in more than a few ways. Some of the ways involve the use of ballot boxes where signed papers are put in them and they are later counted by the teller [4].



Fig.2.2 Ballot boxes used to put ballot papers during voting (made in different materials like cardboard, wood & metal).

Ballot voting was and still is not very reliable as it had a lot of faults, some of which include [5]:

- Prone to rigging by faking number count of the ballot papers.
- Even the use of counting systems sometimes was prone to errors hence falsifying results.
- Security of the system was easily manipulated by tampering of ballot papers after vote ends.
- It is environmental unfriendly as if vote is undertaken for millions of people this means millions of ballot papers and ballot boxes had to be created and printed.
- Storage of the boxes after vote had to be an issue.
- It is costly for larger vote processes.
- It had accessibility issues as it did not cater for the visually impaired or the illiterate as they could not read or write.

In the late 1900's voting slowly became computerized in developed regions with the optical scan ballots, making it easier to vote and eliminate some of the issues that were faced with other voting systems, like storage and environmental friendliness as well as reducing chances or rigging though not eliminating them [5]. By the late 1900's computerized voting had advanced to 2 different types of voting, namely I-voting and E-voting.

As use of the Internet spread rapidly in the 1990s and early 21st century, voters would cast their choices from any computer connected to the Internet—including from their

home, this being I-voting[9]. Use of the Internet has been at times massively disrupted by denial-of-service attacks, this has led to people either choosing the traditional way of voting or migrating to E-voting system as the internet poses risks to the credibility of the votes.

Due to security and access concerns, most large-scale electronic voting is currently held in designated precincts using special-purpose machines. This type of voting mechanism is referred to as E-voting. There are two major types of e-voting equipment: direct recording electronic (DRE) machines and optical scanning machines[13].



Fig.2.3. DRE voting machine(E-voting)

DRE(direct-recording electronic) machines record votes by means of a ballot display provided with mechanical or electro-optical components that can be activated by the voter. These are typically buttons or a touchscreen; and they process data using a computer program to record voting data and ballot images in memory components. After the election, it produces a tabulation of the voting data stored in a removable memory component and as printed copy. The system may also provide a means for transmitting individual ballots or vote totals to a central location for consolidating and reporting results from precincts at the central location. The device started to be massively used in 1996 in Brazil where 100% of the elections voting system is carried out using machines[14].

In some optical scanning systems the voter fills out a paper ballot and inserts it into an electronic scanning device. Scanners can reject improperly marked ballots, allowing the voter to start over, thereby reducing discarded votes[13].



Fig.2.4 Optical-scanner voting machine

In other optical scanning systems voters compose their votes on a computer screen. Once a ballot is completed, the computer prints an optical scanning ballot. The voter verifies the ballot and then inserts it in another device that scans and tabulates the vote[11]. Both E-voting and I-voting systems are considered electronic digital voting systems.

A DRE machine under E-voting systems are a type of machine that is composed by counters, registers and micro-controllers in interconnection with other components. Registers are small, fast storage devices in digital electronics that temporarily hold data, instructions, or addresses for processing within a CPU or digital circuit, whilst a counter is a device which stores (and sometimes displays) the number of times a particular event or process has occurred[15][16]. In our proposed solution a typical but miniature direct-recording electronic(DRE) machine is to be designed with technical advancements to mitigate issues already being faced by already existing voting systems.

2.2 Justification for the proposed voting system project

The digital system comprising of registers and counters has been chosen because it poses major advantages in the voting process which include:

- Implementation of registers and counters ensures that results are displayed in real-time thereby saving time for manual counting as imposed by other methods
- The chosen system minimizes human interaction in voter counting, which therefore means reduced cases of rigging as some may be biased towards a certain individual
- It is cost effective as it reduces expenses incurred in materials like ballot papers since other individuals used to mark more than one individual on the ballot paper which made counting even more difficult
- Also ensures fairness in voting as it won't be biased towards one individual

2.3 Conclusion

Based on the research carried out, the digital voting system is a more secure and accurate method for vote fairness over the other existing methods of voting. The most compelling favour of digital voting over the other methods is convenience, as it gives much easy ways for voters such as saving their valuable time and simple procedures.

Chapter 3: System Design

3.1 Introduction

This chapter presents the overall design of the Digital Voting System. It discusses the architectural, functional, technical and logical design of the voting system. Hardware component selection, system block diagrams, and circuit schematics used to realize the voting process digitally were undergone and done. The goal of the chapter is to present the architectural and functional blueprint of the voting system so as to produce a secure, scalable, and reliable design of an electoral system.

3.2 System Overview

The system's main purpose is to create a safe, secure, efficient, and transparent electronic means of voting. It has to minimize most of the issues recently being faced by existing technologies and methodologies.

3.2.1 Specifications & Functionalities

The desired digital voting system should have the following specifications & functionalities:

- Should allow users to cast votes.
- Should show voting count in real time.
- Should display results in real time upon request.
- Should account for a single vote cast per person.
- Should have an easy user-interface to make the system user friendly.

3.2.2 System requirements

Input unit	Control unit	Counting unit	Display unit
Vote switches as buttons for user to system interaction	Microcontroller for voting system management	Binary counters and registers	Liquid crystal display

Hardware requirements

- Arduino UNO

- Male to male and female to male connection cables
- Buzzer
- Light emitting diode
- Liquid crystal display unit 12c
- Radio Frequency scanner
- Radio Frequency Identification Cards

After further research an Arduino Uno circuit was chosen as our main Operations unit as it encompasses all our desired main components like the counters and registers. The Arduino Uno R3 is a popular microcontroller board used for electronics projects, learning programming, and prototyping, featuring an ATmega328P microcontroller which encompasses 3 counters and several types of registers [17].



Fig.3.1. Aduino Uno R3 board

Software requirements

- Proteus for circuit simulation
- Tinker CAD for circuit construction (ease of use)
- Confluence for flowcharts design
- Lucid chart for data flow diagram construction

3.3 System's software design

This section presents the software design aspect of the Digital Voting System. It outlines the logical flow, algorithms, and control routines that govern how the system

processes voting operations. The software design defines how the system detects user inputs, confirms each vote, updates and stores vote counts, and displays the results on the output devices. *Flowcharts, data flow diagram, case diagrams* and *pseudocode* are used to illustrate the programme structure and logic sequence.

3.3.1 Pseudocode

Pseudocode is a human-readable description of what a programme does. It uses plain English mixed with programming-like structure to outline the steps of an algorithm.

Description of programme

Allow candidates to vote for 3 candidates with Electoral commission using counter to register and shouldn't store results.

Start

Display >> ZEC <<

Display participating parties ["ZANU" "MDC" "CCC"]

Display Number of Votes.

Declare 3 voting variables V1, V2, V3.

Initialize variables to zero.

Initialize RFID to Read ID

```
While (CardID! = officer ID) {  
    Display "Access Denied"  
}  
Display "Session started"
```

set session timer = 0

```
While session time < 54 {  
    If Vote1 = true  
        V1++ [add vote 3+2]  
    else if Vote2 = true {  
        V2++    }  
    else if (Vote3 = true) {  
        V3++}
```

```

Else {
    Display "No votes yet".
} }

End session

IF Vote count = true then {
    Vote = V1 + V2 + V3 compared
    Display winner}
else if there are ties {
    Display Tie
}
else {
    Display No votes.
}
}

END

```

3.3.2 High-level Architectural design

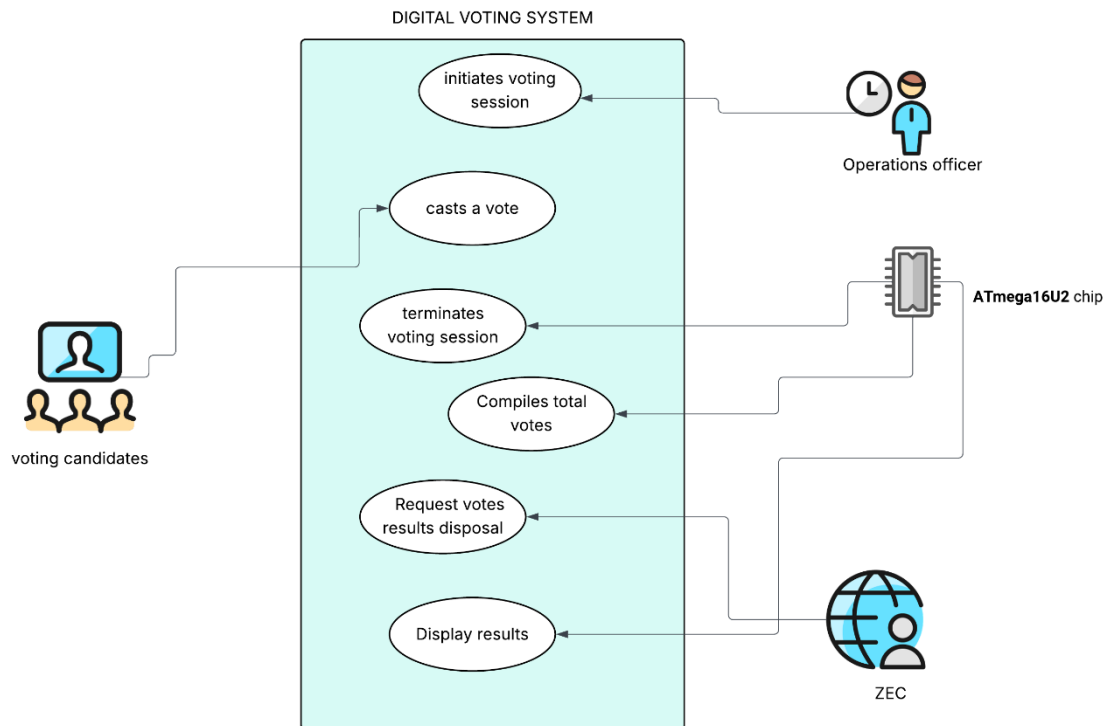


Fig.3.1. High level architecture case diagram

This High-Level Architecture Diagram serves as a visual blueprint that communicates the overall structure and interaction of major components in the system. In this projects context of the voting system , it shows how the individual system entities interact through out the system.

3.3.3 Data Flow Diagram

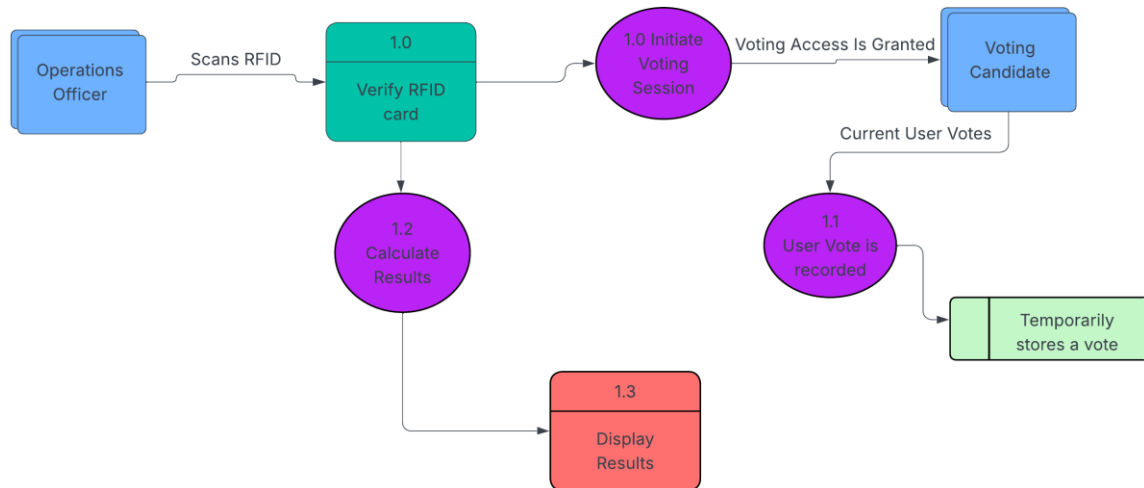


Fig.3.2. Data flow Diagram of the whole system

The Data Flow Diagram (DFD) serves as a powerful visual tool in the system design to illustrate how data moves through it, making it essential for understanding, analyzing, and communicating the system's functionality.

3.3.4 System Flow Charts

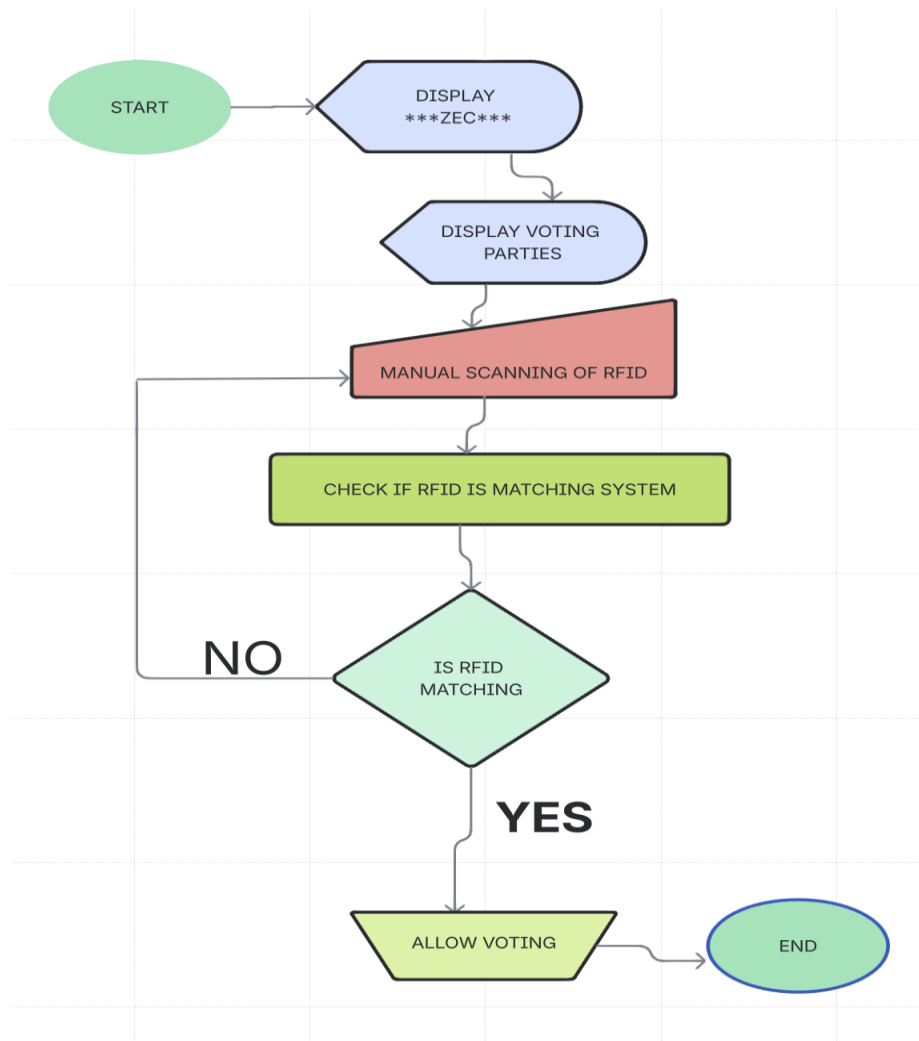


Fig.3.3. Voting process flowchart

A Voting Process Flowchart serves as a visual representation of the step-by-step sequence involved in casting a vote within the system. It translates the logic of the voting process in the system into simple understandable blocks of minute information.

Voting process and outcome evaluation logic flow chart

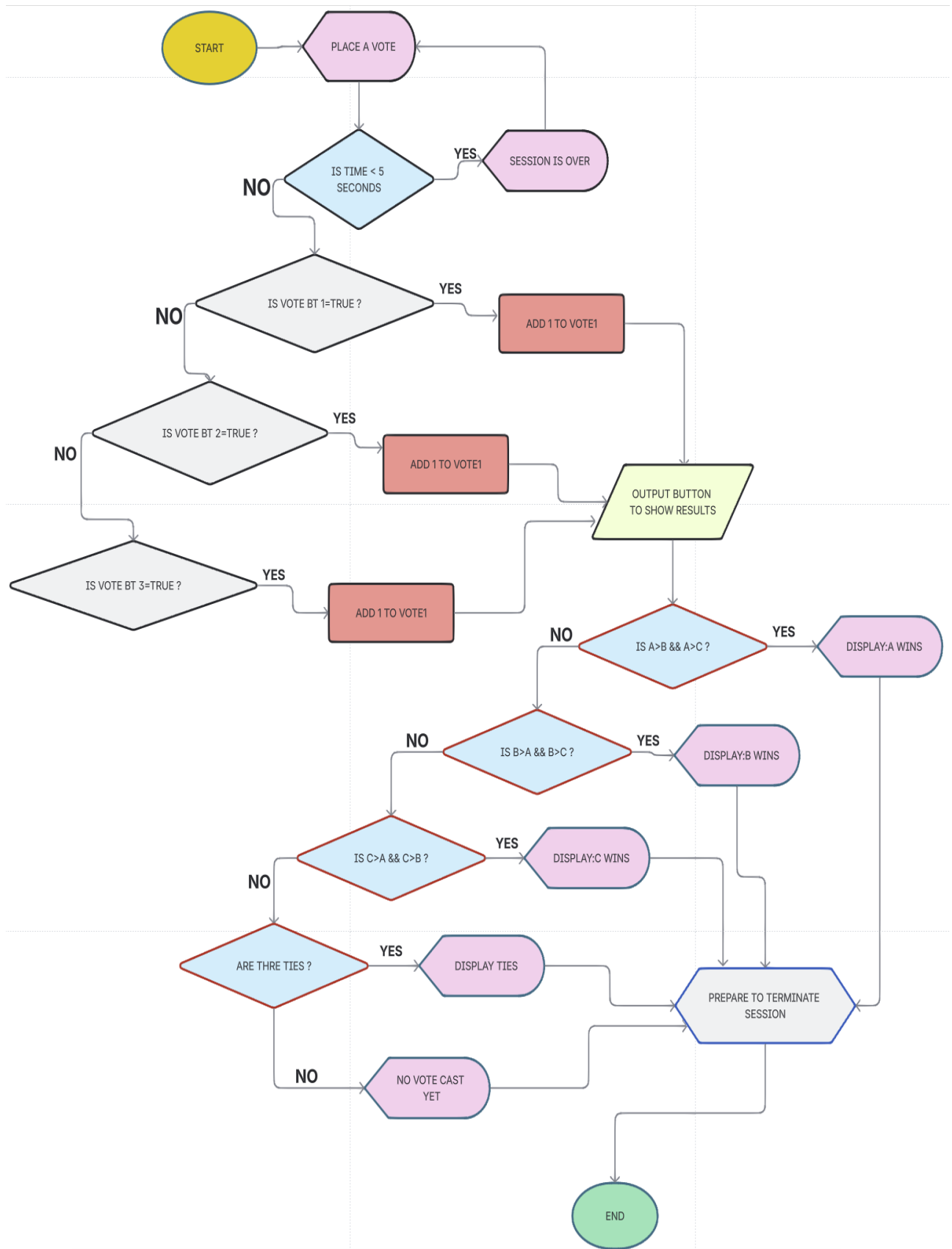


Fig 3.4. Software Voting Process flow chart

This flowchart serves as a simple block visual presentation of the systematic software digitally being inputted and output as the voting process is underway. It represents all complex logic into simple understandable diagrams of information.

Vote count system flow chart

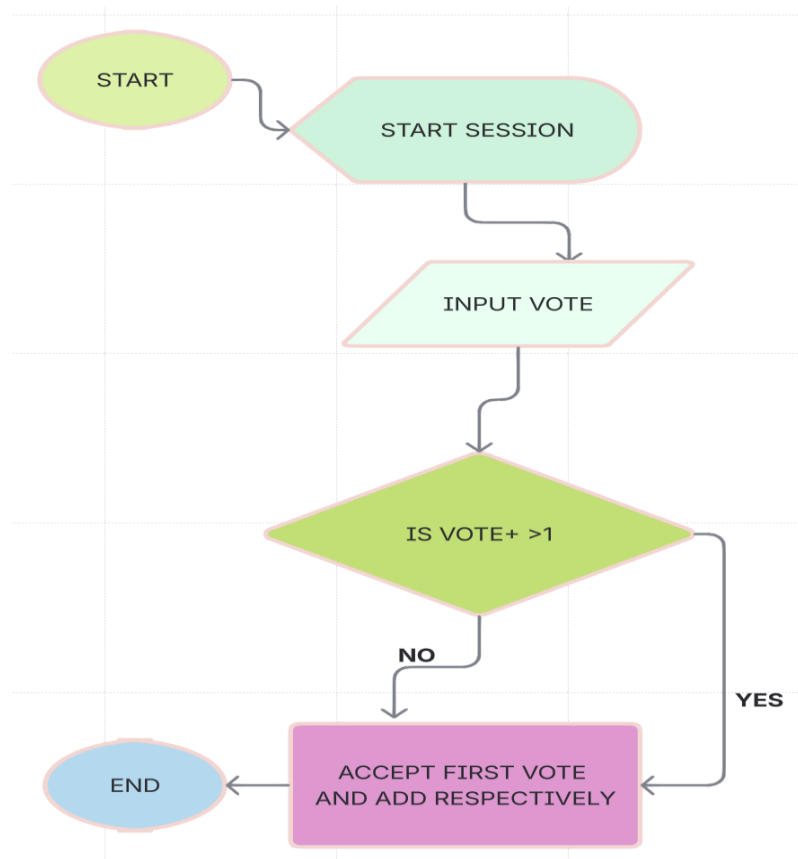


Fig.3.5. Vote count system flowchart

This flowchart illustrates the operational logic of our proposed secure digital voting system designed for the Institution. It begins with system initialization and access control via RFID authentication, ensuring only authorized operations officers can initiate a voting session and view results also. Once access is granted, the session timer activates and voters cast their votes for one of three running candidates or Parties.

Each vote is validated and recorded in real-time. The system continuously monitors the session duration, which is 5 seconds and upon completion, it compiles the total votes, compares party or candidate tallies, and determines the winner. In cases of equal vote counts, a tie is declared; if no votes are recorded, the system flags the session as “No

votes cast yet”. The flowchart emphasizes transparency, security, and automated decision-making throughout the voting life cycle.

3.3.5 Program code


```

1 #include <SPI.h>
2 #include <Wire.h>
3 #include <MLX90.h>
4 #include <LiquidCrystal_I2C.h>
5
6 #define RS_PIN 10
7 #define RW_PIN 9
8
9 #define OFFER_PIN 8
10 #define VOTE_PIN 7
11
12 #define LEFT_BTN 3 // Left button
13 #define RIGHT_BTN 4 // Right button
14 #define CONFIRM_BTN 5 // Confirm vote
15 #define RAZER_BTN 6 // Razer for Address Denied
16 #define SESSION_LED 18
17 #define GREEN_LED 20
18
19 int voted = 0;
20 int voted2 = 0;
21 int voted3 = 0;
22
23 // Session duration in milliseconds (10 seconds)
24 const unsigned long SESSION_DURATION = 10000;
25
26 // Authorize UID
27 byte offerUID[] = {0x23, 0x82, 0x0A, 0x14}; // starts voting session
28 byte offerUID2[] = {0x23, 0x82, 0x0A, 0x01}; // starts voting session
29 byte resultUID[] = {0x23, 0x82, 0x0A, 0x00}; // show results
30
31 bool authorize(byte *uid, byte *storedUID, byte uidSize) {
32   if (uidSize != 4) return false;
33   for (byte i = 0; i < 4; i++) {
34     if (uid[i] != storedUID[i]) return false;
35   }
36   return true;
37 }
38
39 void setup() {
40   SPI.begin();
41   pinMode(RS_PIN, OUTPUT);
42   pinMode(RW_PIN, OUTPUT);
43   pinMode(CONFIRM_BTN, INPUT_PULLUP);
44   pinMode(RAZER_BTN, OUTPUT);
45   pinMode(SESSION_LED, OUTPUT);
46   pinMode(GREEN_LED, OUTPUT);
47 }
48
49 void loop() {
50   // Display busy message
51   lcd.setCursor(0, 0); lcd.print("ZAMU MDC CCC");
52
53   // Scrolling slow message on second line
54   static unsigned long lastScroll = 0;
55   String message = "Voting is a Right and a Responsibility, your vote is your secret, so participate confidently in democracy.";
56   if (millis() - lastScroll > 300) { // scroll speed
57     lcd.setCursor(0, 1);
58     lcd.print(message.substring(scrollPos, scrollPos + 16));
59     scrollPos++;
60     if (scrollPos > message.length() - 16) scrollPos = 0;
61     lastScroll = millis();
62   }
63
64   // PID name
65   if (PIDS.PIDC_IsNewCardPresent() && PIDS.PIDC_IsNewCardSerial()) {
66     if (authenticate(rfid.uid.uidByte, offerUID, rfid.uid.size)) {
67       // Start voting session
68       lcd.clear();
69       lcd.print("Session Started");
70       digitalWrite(SESSION_LED, HIGH);
71       digitalWrite(GREEN_LED, HIGH);
72       delay(2000);
73
74       int selectedParty = 0; // 0=ZAMU, 1=MDC, 2=CCC
75       bool voteCast = false;
76       unsigned long startTime = millis();
77
78       while (millis() - startTime < SESSION_DURATION && !voteCast) {
79         lcd.clear();
80         lcd.setCursor(0, 0); lcd.print("ZAMU MDC CCC");
81
82         // Arrow alignment
83         switch (selectedParty) {
84           case 0: lcd.setCursor(2, 1); lcd.print("<"); break; // ZAMU
85           case 1: lcd.setCursor(1, 1); lcd.print("<"); break; // MDC
86           case 2: lcd.setCursor(15, 1); lcd.print("<"); break; // CCC
87         }
88         delay(100);
89
90         if (digitalRead(LEFT_BTN) == LOW) {
91           selectedParty = (selectedParty + 1) % 3;
92           delay(100);
93         }
94         if (digitalRead(RIGHT_BTN) == LOW) {
95           selectedParty = (selectedParty - 1) % 3;
96           delay(100);
97         }
98         if (digitalRead(CONFIRM_BTN) == LOW) {
99           voteCast = true;
100         }
101
102         lcd.clear();
103         if (voteCast) {
104           switch (selectedParty) {
105             case 0: voteCast++; lcd.print("Vote: ZAMU"); break;
106             case 1: voteCast++; lcd.print("Vote: MDC"); break;
107             case 2: voteCast++; lcd.print("Vote: CCC"); break;
108           }
109         } else {
110           lcd.print("No Vote Cast");
111         }
112         delay(2000);
113         lcd.clear();
114         digitalWrite(SESSION_LED, LOW);
115         digitalWrite(GREEN_LED, LOW);
116       }
117
118       // Show results
119       lcd.clear();
120       int totalVotes = voted + voted2 + voted3;
121       if (totalVotes == 0) {
122         lcd.print("No votes yet");
123         delay(2000);
124         lcd.clear();
125       } else {
126         if (voted > voted2 && voted > voted3) {
127           lcd.print("ZAMU wins");
128         } else if (voted2 > voted && voted2 > voted3) {
129           lcd.print("MDC wins");
130         } else if (voted3 > voted && voted3 > voted2) {
131           lcd.print("CCC wins");
132         } else {
133           String tieMessage = "Tie: ";
134           if (voted == voted2 && voted > voted3) tieMessage += "ZAMU & MDC";
135           else if (voted == voted2 && voted2 > voted3) tieMessage += "ZAMU & CCC";
136           else if (voted == voted3 && voted3 > voted2) tieMessage += "MDC & CCC";
137           else if (voted2 == voted3 && voted2 > voted) tieMessage += "ZAMU MDC CCC";
138           else tieMessage += "Tie/Draw";
139           lcd.print(tieMessage);
140           delay(2000);
141           lcd.setCursor(0, 1);
142           lcd.print("Z"); lcd.print(voted);
143           lcd.print("M"); lcd.print(voted2);
144           lcd.print("C"); lcd.print(voted3);
145           delay(2000);
146           lcd.clear();
147           // Reset votes after showing results
148           voted = 0; voted2 = 0; voted3 = 0;
149         }
150       }
151
152       // Deauthorize card
153       lcd.clear();
154       lcd.print("Access Denied");
155       for (int i = 0; i < 3; i++) {
156         tone(RAZER_PIN, 800, 100);
157         delay(500);
158         tone(RAZER_PIN, 300, 100);
159         delay(500);
160       }
161       delay(2000);
162       lcd.clear();
163       PIDS.PIDC_Halt();
164       PIDS.PIDC_StopCrypto1();
165     }
166   }
167 }

```

Fig.3.6. Program code

- Comments have been included explaining crucial processes in the code.

3.4 System's Hardware Design

The hardware design of the voting system encompasses the physical infrastructure required to support, secure, make reliable, and efficient electoral operations. It integrates RFID authentication devices, push button voting terminals, secure servers, and networking components to facilitate seamless interaction between voters and the system. Each hardware element is selected to ensure data integrity, minimize latency, and uphold the confidentiality of the voting process. The design prioritizes tamper resistance, scalability across polling stations, and compatibility with the system's software architecture.

3.4.1 Circuit schematic

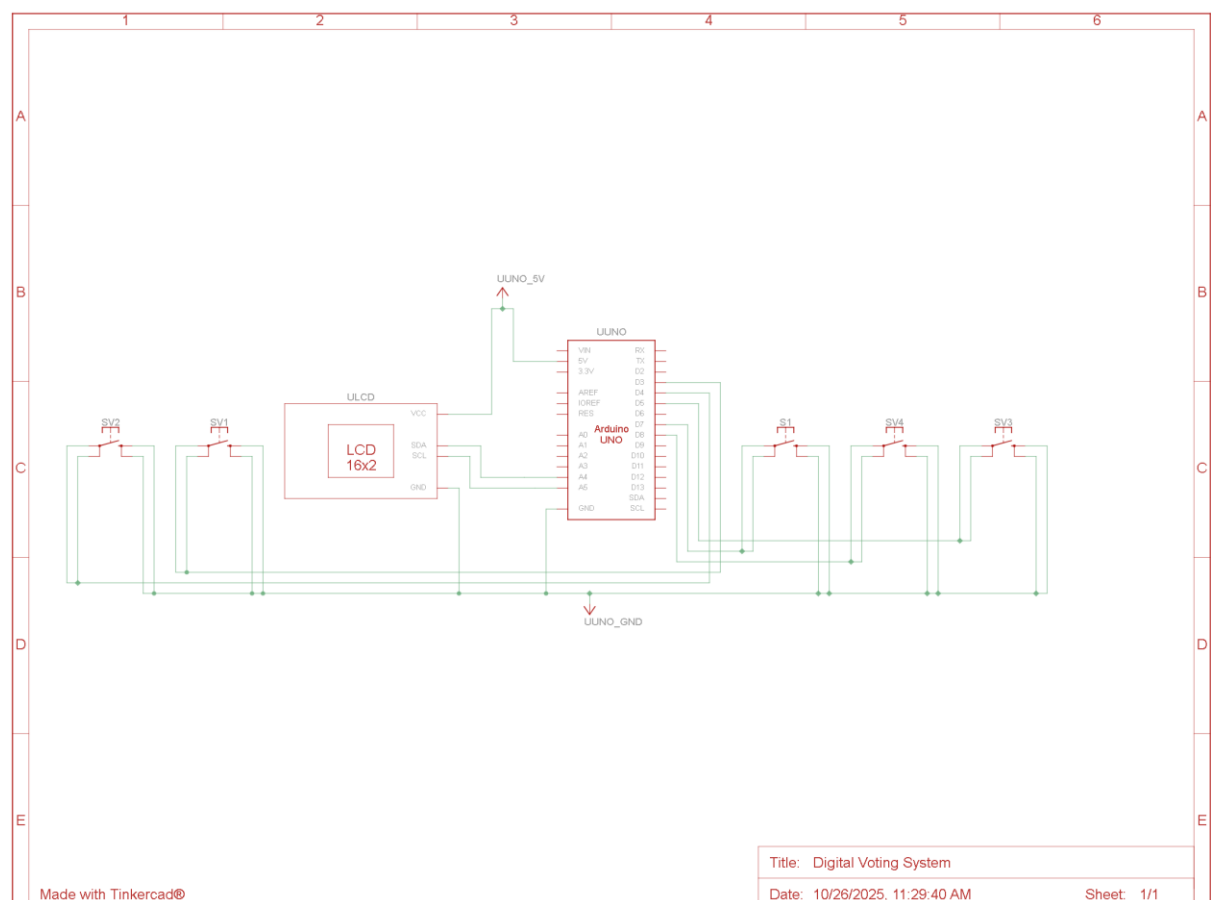


Fig.3.7. Voting system hardware schematic diagram (made with Tinker CAD)

A **schematic circuit diagram** in hardware design serves as a detailed visual representation of the **electrical connections and components** within a system. In the context of this voting system, its purpose is both technical and communicative as it shows the interconnections between the electronic components, helps guide for implementation during production as well as provide a permanent layout of the hardware design structure.

Component List		
Name	Quantity	Component
Ulcd	1	PCF8574-based, 32 (0x20) LCD 16 x 2 (I2C)
Uuno	1	Arduino Uno R3
Sv1 Sv2 Sv3 Sv4 S1	5	Pushbutton

Fig.3.8. Electronic component list on Schematic diagram

3.4.2 Circuit diagram

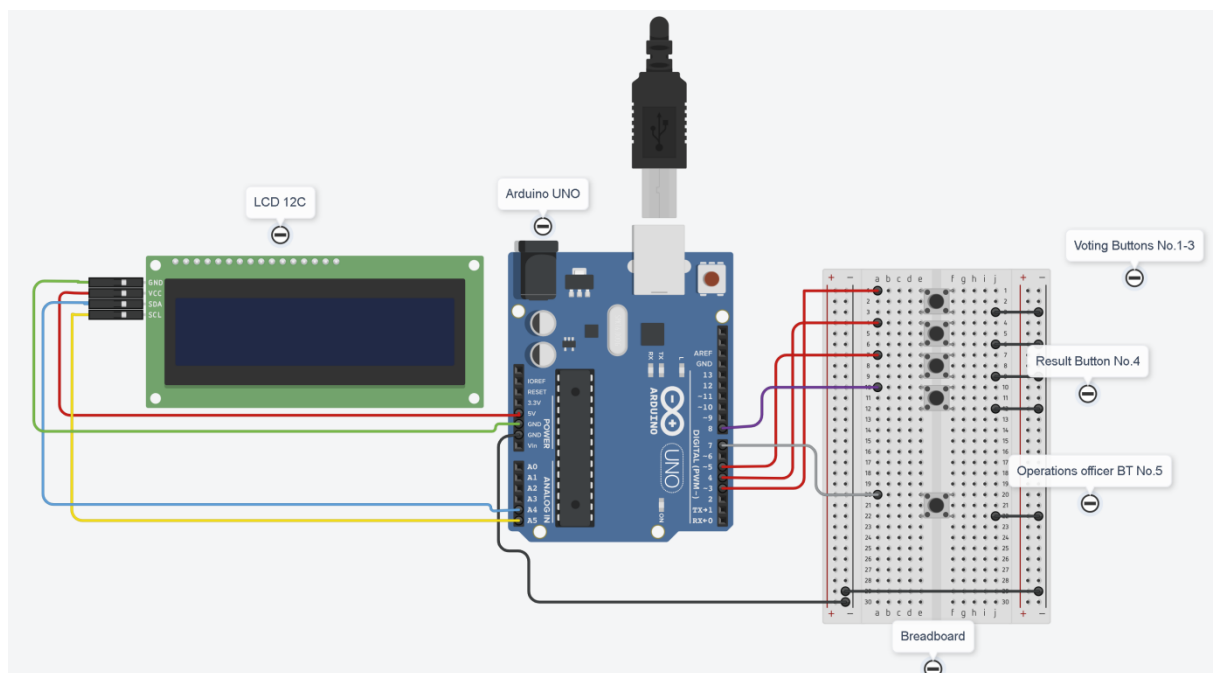


Fig.3.9. Circuit diagram of system hardware (made with Tinker CAD)

***N.B. In place of Operations officer button No.5 there is an RFID scanner.
A buzzer is to be at the top of the breadboard in interconnection with the circuit.***

Hardware components

component	Description
Arduino Uno	Acts as the central processing unit. It reads inputs (RFID, buttons), processes logic, and controls outputs (LCD, buzzer, LED).
RFID Reader (RC522)	Used for voter authentication. The operations officer scans the RFID to grant a vote session.
RFID Tags	Represent individual voters. Each tag contains a unique ID linked to a voter record.
16x2 LCD Display	Provides real-time feedback to the user. Displays prompts, authentication status, and voting confirmation.
Push Buttons	Used for vote selection (candidate1, candidate 2 or candidate 3) and system navigation like to confirm a vote.
Buzzer	Gives audible feedback for actions like invalid tag, or vote confirmation
LED Indicators	Visual cues for system status green to indicate the start and the end of a voting session.
Power Supply	Provides stable voltage to the Arduino and peripherals. Can be USB, battery, or regulated adapter rated 5V.
Breadboard & Jumper Wires	Used for prototyping and connecting components without soldering.

N.B. On the above shown Schematic and circuit diagram there is the absence of some of the noted electronic components in the table above such as buzzer, RFID components and LED indicators due to the absence of the digital components on the software's used for the designing of these diagrams, but are surely to be a part of the prototype of the system.

3.4.3 Justification of choice of components

- Arduino Uno is affordable and easily implements the projects requirement which is to use registers and counters which are encompassed by the ATmega chip which is basically a combination of digital registers and counters.
- RFID to ensure secure contact-less voting session initiation and also implement restriction of unauthorized voting, multiple voting and viewing of results.
- LCD (liquid crystal display) improves visibility and guidance to users for efficient communication with the system.
- Buttons offer simple tactile navigation of the system.

- Buzzer and LED provide multimodal feedback, enhancing accessibility and clarity.

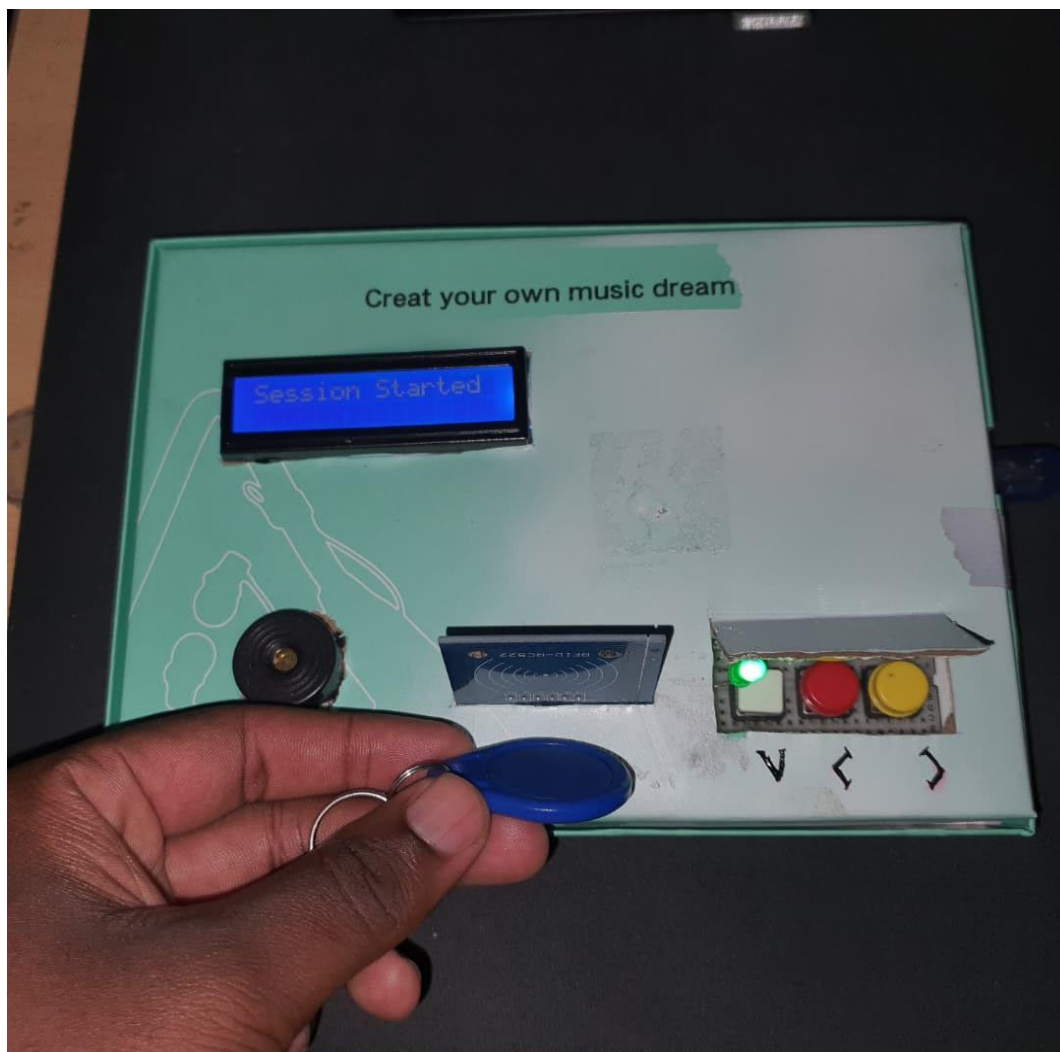
Chapter 4:

This chapter is going to elaborate on the testing done of the digital voting system implementing the use of registers in the digital world. The aim was to design and simulate a digital voting system using counters, registers and display logic. The system was to allow users to cast votes for three candidates, tally votes in real time, and display the results using digital output devices and there no memory storage required.

Normal testing

Test case 1: scanning the correct RFID card to start a voting session

Entity	Response	Results
Registered RFID card	Started a 10sec session	A new voting session was initiated



Test case 2: Voting for a specific candidate

Entity	Response	Results
Anonymous voter	Vote accepted	Voting successful



While evaluating the digital voting system, certain test scenarios could not be fully documented through written records and pictures alone. These include cases where the system's behavior is best observed in real time, such as the physical interaction between the RFID card and the reader, the immediate response of the LCD interface, and the security authentication mechanism after repeated invalid scans. Since these actions involve hardware responses and user interaction, they require physical witnessing to validate their correctness, while digital output responses can be screenshot and recorded, their authenticity is only observed through direct physical observation.

The test has been listed below:

Extreme test:

Test case 1: Invalid card scanned

Entity	Response	Results
Anonymous RFID card (bank card)	Text display: invalid card Buzzer beeps with a tone	No vote session has been initiated

Test case 2: Delayed response

In this test case the developers took the role of the anonymous voter and the other as a security officer and a voting session was initiated.

Entity	Response	Results
Anonymous voter	No vote cast	Returns to Home screen

Test case 3: Validate results where there are no vote casts yet

In this case the developers took the role of the higher tech operations offices, thus here the results are being checked to see the outcome

Entity	Response	Results
TOO (Tech Operations Officers)	No vote cast yet	Resets the system and returns to home screen

Abnormal test:

Test case : Trying to vote without any session initiated by the security officer

In this abnormal test case, the developer tried to vote on the system while he wasn't given access to vote thus the system remained display the home screen without any response.

Entity	Response	Results
Anonymous voter	No response	System remained at the home screen

Chapter 5

Conclusion:

Summary of Objectives

The primary objective of this project was to design, implement, and validate a digital voting system that reliably records votes, enforces role-based access, and resists common manipulation attempts.

Key Achievements

Functional Delivery

The system successfully implements voter authentication, vote capture, and local temporal result storage. Test cases for valid and invalid RFID scans, officer overrides, and vote tallying executed as expected.

Usability

The user interface on the LCD provides clear prompts and feedback for voters and election officers, reducing the likelihood of user error during voting.

Technical Challenges and Developer Response

Code Complexity and Security Hardening

Writing firmware that covers all edge cases and eliminates security loopholes proved challenging. As developers, we iteratively refactored authentication and logging modules to reduce attack surface and to harden input validation.

Testing Limitations

Some hardware interactions required physical witnessing rather than full digital capture. We documented these tests as physically witnessed and supplemented them with logs where possible.

Security Considerations

Threat Reduction

The system incorporates basic protections against unauthorized access and repeated fraudulent attempts. Cryptographic or networked protections were scoped as future enhancements.

Residual Risks

Given the constraints of the current hardware platform, some risks remain (e.g., physical tampering). These are acknowledged and documented with recommended operational controls e.g. supervised polling stations.

Scalability and Future Development

Cloud Integration Path

The architecture is intentionally modular to allow future expansion into **cloud-based centres**. By adding secure network gateways and authenticated result aggregation, multiple polling centres can be connected and results compiled centrally.

Final Statement of Project Aims

The project aimed to design and simulated a digital voting system implemented with counters, registers, and display logic. The system enables voters to cast ballots for three candidates, performs real-time vote tallying through combinational and sequential logic, and presents results immediately on digital output devices. The architecture deliberately omits persistent memory storage, relying on live counting and display for instantaneous verification of results.

All project objectives were achieved: the digital voting system was designed and simulated using counters and registers implemented on an Arduino Uno, together with display logic to support voting for three candidates, perform real-time tallying, and present instantaneous results on digital output devices without persistent memory.

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